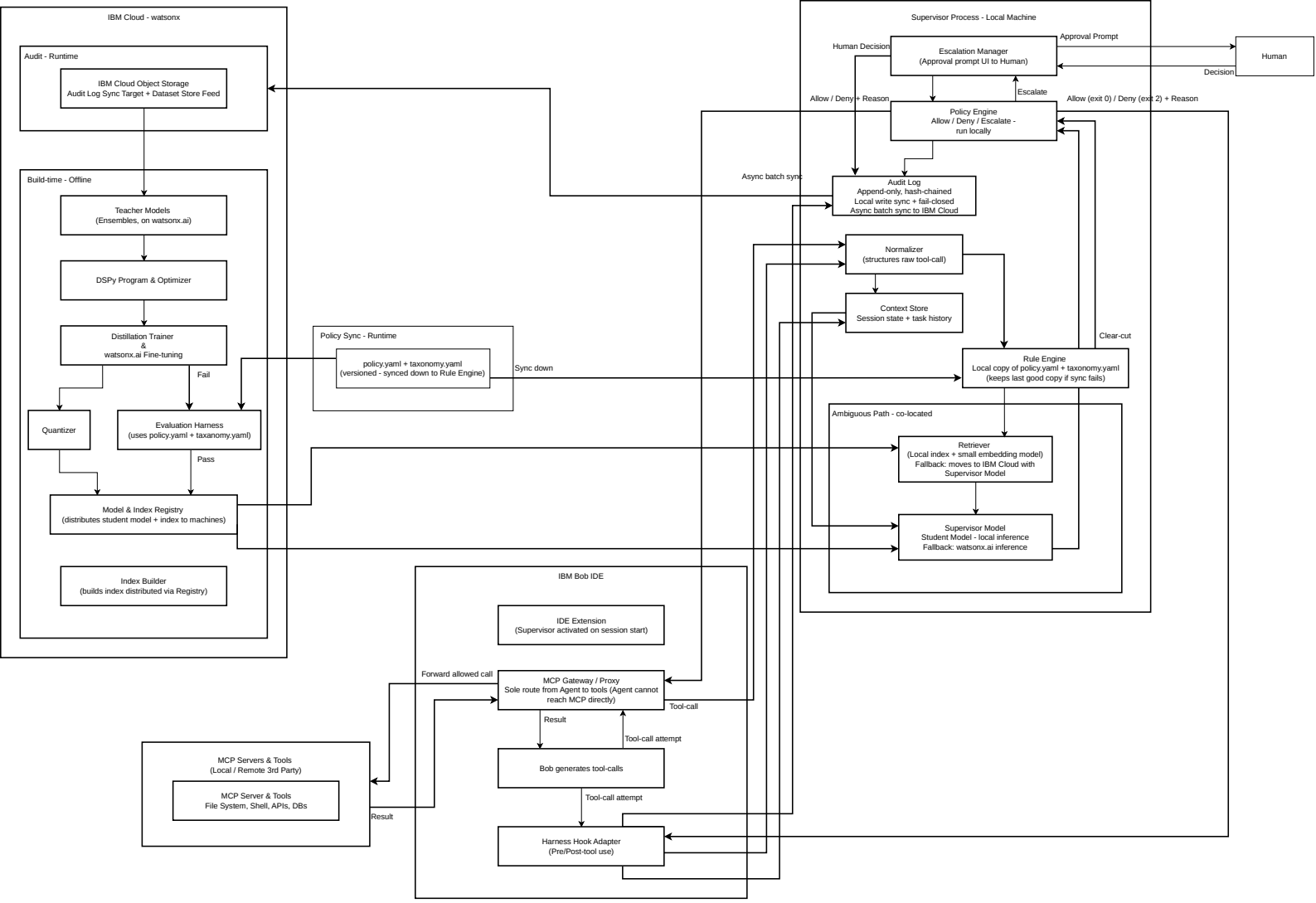
IBM Cloud - watsonx
Audit - Runtime
IBM Cloud Object Storage
Audit Log Sync Target + Dataset Store Feed
Build-time - Offline
Teacher Models
(Ensembles, on watsonx.ai)
DSPy Program & Optimizer
Distillation Trainer
&
watsonx.ai Fine-tuning
Fail
Quantizer
Evaluation Harness
(uses policy.yaml + taxanomy.yaml)
Pass
Model & Index Registry
(distributes student model + index to machines)
Index Builder
(builds index distributed via Registry)
Policy Sync - Runtime
policy.yaml + taxonomy.yaml
(versioned - synced down to Rule Engine)
Sync down
Supervisor Process - Local Machine
Human Decision
Escalation Manager
(Approval prompt UI to Human)
Approval Prompt
Human
Decision
Escalate
Allow / Deny + Reason
Policy Engine
Allow / Deny / Escalate -
run locally
Allow (exit 0) / Deny (exit 2) + Reason
Async batch sync
Audit Log
Append-only, hash-chained
Local write sync + fail-closed
Async batch sync to IBM Cloud
Normalizer
(structures raw tool-call)
Context Store
Session state + task history
Clear-cut
Rule Engine
Local copy of policy.yaml + taxonomy.yaml
(keeps last good copy if sync fails)
Ambiguous Path - co-located
Retriever
(Local index + small embedding model)
Fallback: moves to IBM Cloud with
Supervisor Model
Supervisor Model
Student Model - local inference
Fallback: watsonx.ai inference
IBM Bob IDE
IDE Extension
(Supervisor activated on session start)
Forward allowed call
MCP Gateway / Proxy
Sole route from Agent to tools (Agent cannot
reach MCP directly)
Tool-call
Result
Tool-call attempt
Bob generates tool-calls
Tool-call attempt
Harness Hook Adapter
(Pre/Post-tool use)
MCP Servers & Tools
(Local / Remote 3rd Party)
MCP Server & Tools
File System, Shell, APIs, DBs
Result